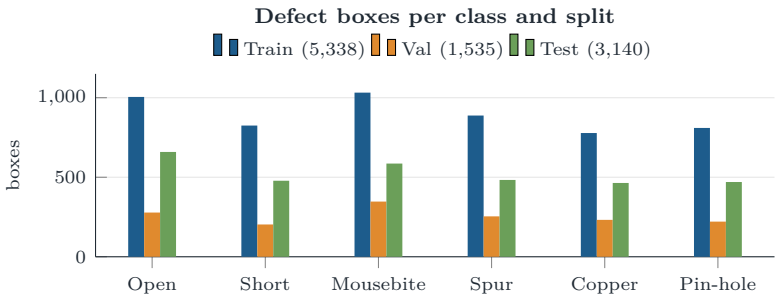
PCB Defect Detection Under Image Misalignment

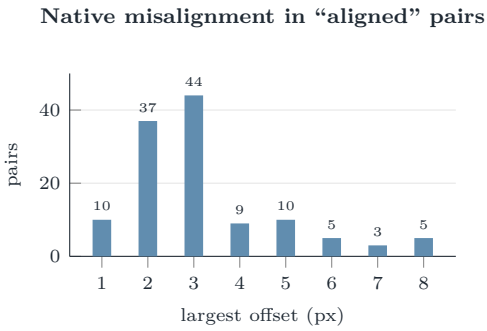
Data Report

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This report describes the data used to test whether a compact learned detector can separate genuine PCB defects from misalignment artefacts: the DeepPCB source, the grouped partition, a quality audit of the raw pairs, the seeded misalignment protocol, and the checks applied to every prepared image and box. All numbers come from the scripts and JSON reports in the project repository.

1,500	10,013	800/200/500	7,000	123	9/9
image pairs	defect boxes	train / val / test	val+test images	natively misaligned	checks passed





1 Source dataset

DeepPCB [1] contains 1,500 pairs of 640×640 near-binary images: a defect-free *template* (reference) and an *inspected* image of the same PCB region, aligned by the authors with template matching. Each pair has 1–15 boxes (mean 6.7) labelled as open, short, mousebite, spur, spurious copper or pin-hole. Defects are small (median 35 × 33 px, about 0.3 % of the image) and classes are close to balanced (largest/smallest ratio 1.33). Most defects were added artificially by the dataset authors, which limits claims about real production defects. The official lists hold 1,000 *trainval* and 500 *test* pairs from 11 source-board *groups*; the group folder is the only board identifier available.

2 Partition

The official 1,000/500 split is kept. About 20 % of trainval is held out for validation *before* any augmentation, grouped by board so that no board appears in both train and validation. Every subset of trainval groups with 17–23 % of the pairs was scored by the L_1 distance between its class proportions and those of all trainval; the best is group `00041` (200 pairs, $L_1 = 0.059$). All shifted variants of a pair inherit the pair's split. Because the official test list was kept, the rule “same board, same partition” cannot hold for test: 168 test pairs come from 7 groups also present in trainval (21 from the validation group). The other 332 come from four groups never seen in trainval (`12000`, `12100`, `12300`, `90100`), so a group-held-out validation set is a reasonable proxy for test.

3 Data-quality audit

- **Format.** All 3,000 images are 640 × 640; 62 images in 57 pairs are 3-channel JPEGs and were converted to single-channel PNG. No label file is empty, malformed or out of bounds.
- **Near-duplicates.** A difference hash over all templates gave 337 candidates, re-checked at pixel level (best offset within ±8 px, Jaccard ≥ 0.9). 40 duplicates were confirmed, all inside one group, which supports group as the board unit. None crosses train/val; three test pairs (`50600065`, `92000114`, `92000118`) duplicate a train pair and are flagged for a with/without comparison.
- **Native misalignment.** An integer search (±8 px) for the offset that minimises template/inspected disagreement finds 123 pairs (8.2 %; 75 train, 8 val, 40 test) where an offset cuts the XOR area by at least 1.5×, typically 2–3 px (chart above, Fig. 1). Their “aligned” condition is therefore already shifted. Offsets are stored per pair, so results can be reported on all pairs and, as a sensitivity check, on the 1,377 clean pairs (460 in test). 17 templates are near-blank.

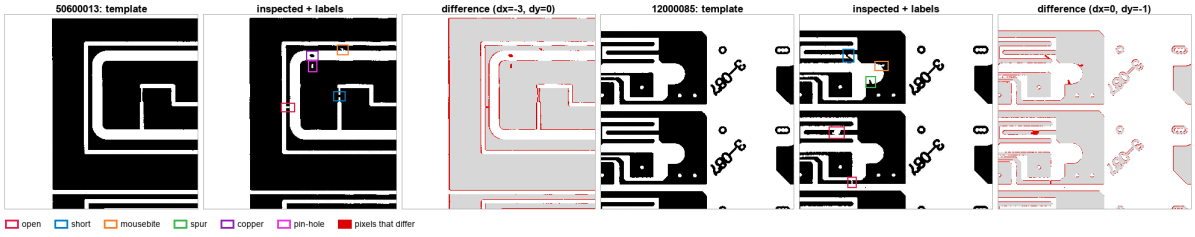


Figure 1: Two pairs supplied as “aligned” but offset by 3 px and 1 px (offset found by the audit). Boxes show the labelled defects by class; red marks pixels that differ between template and inspected image.

4 Controlled misalignment protocol

The reference is never modified; the inspected image and its boxes move together. Each validation and test pair gets ten conditions (Table 1): the pair as supplied (`s0`); integer shifts of 1, 2, 3 and 5 px along x or y with a random sign; and a *residual* condition r (sub-pixel translation of 0.25–1.5 px plus a rotation within $\pm 0.5°$) that whole-pixel registration cannot undo exactly. Integer shifts copy pixels, so they add no interpolation artefacts; r uses bilinear interpolation, and rotated boxes become the axis-aligned envelope of their corners. Parameters come from a generator seeded with (6314, pair ID), so the baseline and the detector read the *same* files. **Training does see misalignment:** the non-augmented run uses the 800 pairs as supplied (75 already natively offset), and the augmented run adds one random 1–5 px shift and one residual copy of every pair (2,400 images, separate seed stream; Section 6).

Table 1: Prepared inspected images.

Split	`s0`	1/2/3/5 px $\times\{x, y\}$	r	Images
Train	800	–	–	800
Val	200	8×200	200	2,000
Test	500	8×500	500	5,000

Borders and truncated defects. Uncovered pixels are edge-replicated inside an 8 px border that evaluation excludes. A box whose *original* position is within 14 px of an edge is marked *ignore* in every condition (258 boxes, 2.6 %): it counts as neither hit nor miss, and a prediction matching it is not a false positive.

5 Verification

Two sets of checks were run: the pipeline's own QA scripts, and an independent audit that re-decodes the raw DeepPCB files with a different library (Pillow/SciPy instead of OpenCV) and does not reuse the pipeline code. The only difference found is one colour JPEG (`44000020`) whose grey conversion differs between decoders on 2 % of pixels; its shifted copies still match its own `s0` exactly.

Check	Result
Pair lists, splits, `boxes.csv` match raw	1500/1500
References identical to raw templates	1500/1500
`s0` identical to raw (independent decoder)	1499/1500
Integer shift = `s0` moved by (dx, dy)	5600/5600
Box count, class, offset, ignore flag	52088/52088
Non-ignored boxes in valid region	41220/41220
Residual images reproduced from parameters	700/700
COCO files consistent with CSV	21/21
YOLO images with labels, zip CRC intact	10200/10200

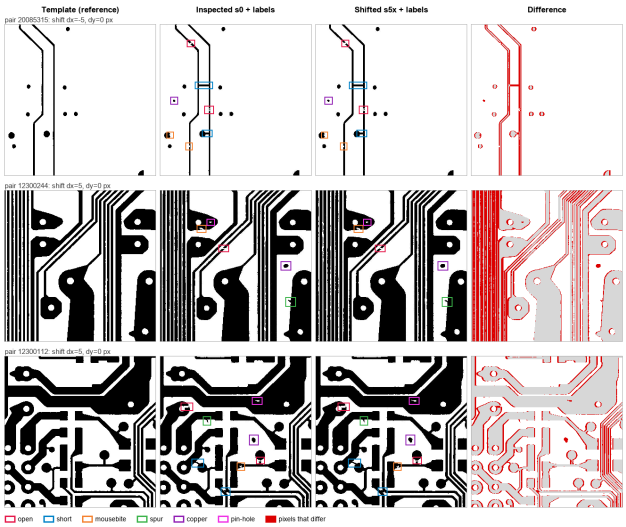


Figure 2: Label check, 5 px shift: template; inspected `s0` with labels; shifted image with shifted labels; difference in red, where every track edge becomes a false difference.

6 Model-ready formats and reproducibility

The master format is CSV: `splits.csv` (pair, group, split, native offset, duplicate flag), `images.csv` (one row per inspected image with dx, dy, rotation) and `labels.csv` (one row per box with class and ignore flag). The same data are exported as 21 COCO files and as a YOLO package where each pair is one 3-channel image (reference, inspected, |difference|), so an RGB-pretrained YOLOv10-Nano takes both images unchanged. The augmented run (3× the images) trains for one third of the epochs, so both runs take the same number of steps. Colour augmentation stays off because it would corrupt the difference channel; YOLO cannot encode ignore boxes, so evaluation reads `labels.csv`. Everything is rebuilt from the untouched DeepPCB download with `scripts/run_all.sh`.

7 Limitations

Most defects are synthetic; validation is a single board group; each pair has one residual variant; 123 pairs are natively misaligned; and 168 test pairs share a board with trainval. These are reported with every result rather than corrected.

References

[1] S. Tang, F. He, X.-L. Huang, and J. Yang, "Online PCB defect detector on a new PCB defect dataset," *arXiv:1902.06197*, 2019.